AI/ML intern DAY 19

AI/ML Internship Final Key Learnings Report
AI-Based Virtual Try-On and Fashion Recommendation System

LOYA PRANAY (pranay8679@gmail.com)

AI/ML Intern

Date: 05-06-2026

---

1. Introduction

The AI/ML Internship at Raritone provided an excellent opportunity to explore the practical applications of Artificial Intelligence, Computer Vision, and Machine Learning in the field of fashion technology. Throughout the internship, I worked on research-oriented tasks related to body segmentation, pose estimation, body landmark detection, garment warping, Virtual Try-On systems, fashion datasets, recommendation systems, and 3D avatar generation.

The internship combined theoretical research with practical exploration, helping me understand how multiple AI technologies work together to create intelligent digital fashion platforms. In addition to technical knowledge, the internship also strengthened my analytical thinking, documentation skills, and ability to study modern AI architectures independently.

---

2. Understanding AI-Powered Virtual Try-On Systems

One of the most valuable learnings during the internship was understanding the complete workflow of AI-based Virtual Try-On systems.

Initially, I viewed Virtual Try-On as a simple image overlay process. However, through detailed research, I learned that realistic garment fitting requires the integration of multiple Artificial Intelligence modules.

The major components studied include:

- Human Detection
- Body Segmentation
- Pose Estimation
- Landmark Detection
- Body Measurement Extraction
- Garment Warping

- Image Blending
- Recommendation Systems
- 3D Avatar Generation

I learned that every module contributes to generating a realistic and personalized digital fitting experience.

---

3. Computer Vision and Pose Estimation

A significant portion of the internship focused on understanding Computer Vision technologies.

I studied how AI systems detect and understand the human body using pose estimation algorithms.

Technologies Explored

- MediaPipe Pose
- OpenPose
- Detectron2
- DeepLabV3
- MediaPipe Selfie Segmentation

Through this research, I gained a clear understanding of how body key points are extracted and how they support garment alignment and body measurement estimation.

I also learned the importance of real-time landmark detection for Virtual Try-On and interactive AI applications.

---

4. Body Landmark Detection and Measurement Extraction

Another important learning outcome was understanding how body landmarks can be used to estimate physical body dimensions.

The measurements researched include:

- Height Estimation
- Shoulder Width
- Waist Width
- Hip Width
- Arm Length

I learned that accurate body measurements play a critical role in improving garment fitting quality and recommendation accuracy.

The internship helped me understand how pose estimation and landmark detection work together to produce these measurements.

---

5. Fashion Datasets and Data Preparation

The internship provided valuable knowledge about the importance of high-quality datasets in Artificial Intelligence projects.

Several publicly available fashion datasets were studied, including:

- DeepFashion
- DeepFashion2
- VITON
- VITON-HD
- Dress Code
- MPV Dataset
- Fashionpedia

In addition to studying the datasets themselves, I learned the importance of:

- Data Collection
- Data Cleaning
- Metadata Generation
- Image Annotation
- Training and Validation Splits

This research demonstrated that high-quality data is one of the most important factors affecting AI model performance.

---

6. Garment Warping and Virtual Try-On Models

One of the most interesting areas explored during the internship was garment warping.

I learned that garments cannot simply be pasted onto a user's image. Instead, they must be transformed to match body shape and posture.

Technologies Studied

- Thin Plate Spline (TPS)
- Geometric Matching Module (GMM)
- Flow-Based Warping
- Affine Transformation

I also researched several modern Virtual Try-On architectures:

- VITON
- CP-VTON
- HR-VITON
- StableVITON
- IDM-VTON

Comparing these models helped me understand the evolution of AI-powered Virtual Try-On systems and the importance of image quality, texture preservation, and realistic garment fitting.

---

7. 3D Avatar Generation and Human Reconstruction

The internship introduced me to the concept of creating digital human avatars using Artificial Intelligence.

Technologies studied include:

- SMPL
- PIFuHD
- ICON
- HumanNeRF
- 3D Gaussian Splatting

I learned how these technologies reconstruct human body geometry from images and how they contribute to future generations of Virtual Try-On platforms and digital fashion experiences.

This research also expanded my understanding of AR, VR, and Metaverse applications.

---

8. AI Recommendation Systems

An important learning outcome was understanding how Artificial Intelligence can personalize shopping experiences.

Recommendation approaches studied include:

- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation Systems
- Deep Learning Recommendation Models

I learned how user preferences, browsing history, and body measurements can be combined to generate personalized outfit suggestions.

---

9. Technical Skills Developed

The internship strengthened several technical skills.

Artificial Intelligence

- Machine Learning Research
- Computer Vision
- Deep Learning Concepts

Programming and Tools

- Python
- OpenCV
- MediaPipe
- GitHub
- Visual Studio Code

Fashion AI Technologies

- Virtual Try-On Systems
- Body Segmentation
- Landmark Detection
- Garment Warping
- 3D Avatar Generation

---

10. Research and Professional Skills Developed

Apart from technical knowledge, the internship significantly improved my research and professional abilities.

Skills Developed

- Technical Research
- Comparative Analysis
- Documentation Writing
- Independent Learning
- Critical Thinking

- Problem Solving
- Project Organization
- GitHub Repository Management

I learned how to study research papers, compare AI architectures, document findings, and present technical information in a structured manner.

---

11. Major Takeaways

Some of the most important takeaways from this internship include:

- Understanding the complete AI workflow behind Virtual Try-On systems.
- Learning how multiple AI technologies integrate into a single application.
- Appreciating the importance of high-quality datasets and annotations.
- Understanding body landmark detection and measurement extraction.
- Exploring advanced garment warping techniques.
- Learning the future potential of AI-powered fashion technology.
- Improving research, documentation, and analytical skills.

---

12. Overall Experience

The internship was a valuable learning experience that combined technical research, practical exploration, and professional documentation.

Working on different AI modules helped me understand the complexity of modern intelligent systems and the importance of interdisciplinary knowledge.

The combination of Computer Vision, Machine Learning, and fashion technology provided a unique opportunity to explore real-world AI applications.

The internship also improved my confidence in independently learning and analysing emerging technologies.

---

13. Conclusion

The Raritone AI/ML Internship provided comprehensive exposure to modern Artificial Intelligence technologies and their applications in digital fashion platforms.

Through research and exploration, I developed a strong understanding of Computer Vision, body segmentation, pose estimation, body measurement extraction, fashion datasets, garment warping, Virtual Try-On systems, recommendation engines, and 3D avatar generation.

In addition to strengthening my technical knowledge, the internship enhanced my research methodology, documentation skills, analytical thinking, and independent learning abilities. The experience established a solid foundation for my future studies and career growth in Artificial Intelligence, Machine Learning, and Computer Vision, while also deepening my interest in building intelligent systems that solve real-world problems.